

Tin

50 Sn Mined since 3000 BCE, tin is combined with copper to form bronze, a strong alloy. Bronze played an important role in the development of many civilizations, allowing people to make weapons, tools, utensils, and statues. Tin is also combined with lead or other metals to make solder—a fusible alloy used to join metals together, especially in electronic devices.



Laboratory sample of pure tin

Bronze

Most modern bronze sculptures contain roughly 10 percent tin (the rest is copper). Harder, and more resistant to corrosion than iron, bronze has been an ideal material for sculptors for nearly 5,000 years. Modern applications also include propellers for ships and strings for guitars.



Auguste Rodin's bronze sculpture *The Thinker*, in Stockholm, Sweden

Tin toy train from 1900

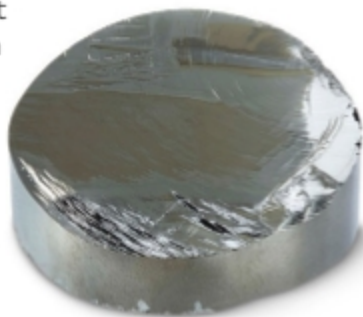


Child's play

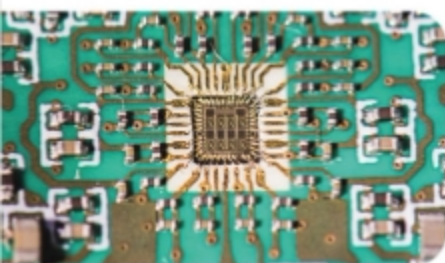
The malleability of tin made it useful in making toys over the last two centuries. Toys were often made with tinplate—a steel plate coated with a thin layer of tin. These days, such toys are collectible items—modern toys are more likely to be made of plastic.

Germanium

32 Ge This brittle semi-metal was discovered in 1886 by the German chemist Clemens Alexander Winkler, and it is named after his home country, Germany. Germanium is an important element in technology, used in the production of high-quality lenses for microscopes and cameras.



Laboratory-refined germanium disc



Computer chips

Like silicon, germanium is a semiconductor, and it can be used to make microchips. It is combined to make silicon-germanium chips that work at high speeds, as shown here.

Flerovium

114 Fl A synthetic element, flerovium was first produced in 1998 when scientists in Russia smashed together plutonium and calcium atoms. This highly radioactive element was given its official name in 2012. It sits in group 14, but scientists are still debating its properties. Some think it might have metallic properties, while others think it may behave more like a noble gas.

Georgy Flyorov

Flerovium is named after the Flerov Laboratory of Nuclear Reactions in Russia, where the element was discovered. The laboratory, in turn, is named after the Russian physicist Georgy Flyorov.

